

Ratios, Proportion & Percentages

ANSWER KEY



Simplifying ratios

- 1 Simplify the ratio 18 : 24.

GCD of 18 and 24 is 6: $18 \div 6 : 24 \div 6 = 3 : 4$.

- 2 Simplify the ratio 45 : 60 : 75.

GCD of 45, 60, 75 is 15: $3 : 4 : 5$.

- 3 Simplify the ratio 36 : 8.

GCD of 36 and 8 is 4: $9 : 2$.

- 4 Express "12 boys to 18 girls" as a ratio in simplest form.

12 : 18, and GCD is 6, so $2 : 3$.

Sharing a quantity in a given ratio

- 5 Share 150 in the ratio 2 : 3.

$2 + 3 = 5$ parts, each part $= 150 \div 5 = 30$. Shares: 60 and 90.

- 6 Share 220 among three people in the ratio 3 : 4 : 4.

$3 + 4 + 4 = 11$ parts, each part $= 220 \div 11 = 20$. Shares: 60, 80, 80.

- 7 A prize of 900 riyals is shared between Sara and Layla in the ratio 5 : 4. Find each share.

$5 + 4 = 9$ parts, each part $= 900 \div 9 = 100$. Sara receives 500 riyals; Layla receives 400 riyals.

- 8 Divide 84 kg of flour in the ratio 1 : 2 : 4.

$1 + 2 + 4 = 7$ parts, each part $= 84 \div 7 = 12$ kg. Shares: 12 kg, 24 kg, 48 kg.

Equivalent ratios and proportions

- 9 If $3 : 5 = x : 20$, find x .

$$\frac{3}{5} = \frac{x}{20}, \text{ so } x = \frac{3 \times 20}{5} = 12.$$

- 10 If $a : 7 = 12 : 21$, find a .

$$\frac{12}{21} = \frac{4}{7}, \text{ so } a = 4.$$

- 11 A map has scale $1 : 50,000$. A distance of 3 cm on the map represents how many meters in reality?

$$3 \times 50,000 = 150,000 \text{ cm} = 1,500 \text{ m}.$$

- 12 Solve for y : $\frac{4}{9} = \frac{y}{63}$.

$$y = \frac{4 \times 63}{9} = 28.$$

Percent of a number

- 13 Find 35% of 240.

$$\frac{35}{100} \times 240 = 84.$$

- 14 Find 12% of 150.

$$\frac{12}{100} \times 150 = 18.$$

- 15 What is 8% of 650?

$$\frac{8}{100} \times 650 = 52.$$

- 16 Find 150% of 40.

$$\frac{150}{100} \times 40 = 60.$$

Converting between fractions, decimals and percentages

- 17 Write $\frac{3}{8}$ as a percent.

$$\frac{3}{8} = 0.375 = 37.5\%.$$

- 18 Write 0.45 as a fraction in lowest terms.

$$0.45 = \frac{45}{100} = \frac{9}{20}.$$

- 19 Write 62.5% as a fraction in lowest terms.

$$62.5\% = \frac{625}{1000} = \frac{5}{8}.$$

- 20 Write $\frac{7}{40}$ as a decimal.

$$\frac{7}{40} = 0.175.$$

Percentage increase and decrease

- 21** Increase 80 by 25%.

$$80 \times 1.25 = 100.$$

- 22** Decrease 150 by 20%.

$$150 \times 0.8 = 120.$$

- 23** A salary of 4,000 riyals increases by 6%. Find the new salary.

$$4,000 \times 1.06 = 4,240 \text{ riyals.}$$

- 24** A population of 2,500 decreases by 12%. Find the new population.

$$2,500 \times 0.88 = 2,200.$$

Finding percentage change and original values

- 25** A price rises from 80 to 92. Find the percentage increase.

$$\text{Change} = 92 - 80 = 12. \text{ Percent increase} = \frac{12}{80} \times 100\% = 15\%.$$

- 26** A quantity falls from 250 to 200. Find the percentage decrease.

$$\text{Change} = 250 - 200 = 50. \text{ Percent decrease} = \frac{50}{250} \times 100\% = 20\%.$$

- 27** After a 20% discount, a jacket costs 96 riyals. Find the original price.

$$\text{Original} \times 0.8 = 96, \text{ so original} = 96 \div 0.8 = 120 \text{ riyals.}$$

- 28** After a 15% increase, a value becomes 138. Find the original value.

$$\text{Original} \times 1.15 = 138, \text{ so original} = 138 \div 1.15 = 120.$$

Successive percentage changes

- 29** A price of 100 rises 10%, then falls 10%. Find the final price.

$$100 \times 1.1 \times 0.9 = 99 \text{ — one riyal less than the start, since the second change applies to a larger base.}$$

- 30** A stock worth 200 riyals rises 20% one year, then rises a further 25% the next year. Find the final value.

$$200 \times 1.2 \times 1.25 = 300 \text{ riyals.}$$

- 31** A quantity decreases by 10%, then decreases by a further 10%. Find the overall percentage decrease from the original amount.

$100 \times 0.9 \times 0.9 = 81$, a fall of 19 from 100 — an overall decrease of 19% (not 20%).

- 32** A price of 400 riyals increases by 5%, then decreases by 5%. Find the final price, and state whether it equals the starting price.

$400 \times 1.05 \times 0.95 = 399$ riyals — one riyal less than the original 400, so it is not equal.

Direct proportion

- 33** y is directly proportional to x , and $y = 15$ when $x = 3$. Find y when $x = 8$.

$y = kx$ with $k = 15 \div 3 = 5$. When $x = 8$: $y = 5 \times 8 = 40$.

- 34** 4 kg of apples cost 28 riyals. Find the cost of 7 kg at the same rate.

Cost per kg = $28 \div 4 = 7$ riyals. Cost of 7 kg = $7 \times 7 = 49$ riyals.

- 35** A car uses 8 liters of fuel to travel 96 km. How far can it travel on 20 liters at the same rate?

Rate = $96 \div 8 = 12$ km per liter. Distance = $20 \times 12 = 240$ km.

- 36** 6 identical pens cost 27 riyals. Find the cost of 10 pens.

Cost per pen = $27 \div 6 = 4.5$ riyals. Cost of 10 pens = $10 \times 4.5 = 45$ riyals.

Inverse proportion

- 37** 4 workers can complete a job in 15 days. How many days would 6 workers take, working at the same rate?

Total work = $4 \times 15 = 60$ worker-days. With 6 workers: $60 \div 6 = 10$ days.

- 38** A car covers a fixed distance in 5 hours at 60 km/h. How long would the same trip take at 75 km/h?

Distance = $60 \times 5 = 300$ km. Time at 75 km/h: $300 \div 75 = 4$ hours.

- 39** 8 taps fill a pool in 12 hours. How long would 3 taps take, running at the same rate?

Total work = $8 \times 12 = 96$ tap-hours. With 3 taps: $96 \div 3 = 32$ hours.

Best value and unit price comparisons

- 40** A 3 kg bag of rice costs 27 riyals, and a 5 kg bag costs 42.50 riyals. Which is the better value per kilogram?

3 kg bag: $27 \div 3 = 9$ riyals/kg. 5 kg bag: $42.5 \div 5 = 8.5$ riyals/kg. The 5 kg bag is better value.

- 41** Store A sells 6 pens for 15 riyals; Store B sells 8 pens for 19.20 riyals. Which store offers the lower price per pen?

Store A: $15 \div 6 = 2.5$ riyals/pen. Store B: $19.2 \div 8 = 2.4$ riyals/pen. Store B has the lower price per pen.

- 42** A 2-liter bottle of juice costs 14 riyals. Find the price per 100 mL.

2 liters = 20 portions of 100 mL, so price per 100 mL = $14 \div 20 = 0.7$ riyals.

- 43** Which is the better deal: 12 eggs for 9 riyals, or 18 eggs for 12.60 riyals?

12 eggs: $9 \div 12 = 0.75$ riyals/egg. 18 eggs: $12.6 \div 18 = 0.70$ riyals/egg. The 18-egg pack is the better deal.

Word problems

- 44** A recipe requires flour and sugar in the ratio 5 : 2. If 350 g of flour is used, how much sugar is needed?

Scale factor = $350 \div 5 = 70$. Sugar = $2 \times 70 = 140$ g.

- 45** A shop raises the price of a shirt from 60 to 75 riyals, then during a sale reduces the new price by 20%. Find the sale price.

New price after the sale: $75 \times 0.8 = 60$ riyals — back to the original price.

- 46** In a class of 40 students, 60% are girls. Then 4 more girls join the class. What percent of the class is now girls, to the nearest percent?

Girls initially = $0.6 \times 40 = 24$; boys = 16. After 4 more girls: 28 girls out of 44 students.
 $\frac{28}{44} \times 100\% \approx 64\%$.

- 47** A dealership reduces the price of a car by 15% to 25,500 riyals. Find the original price.

Original $\times 0.85 = 25,500$, so original = $25,500 \div 0.85 = 30,000$ riyals.

- 48** Ahmed invests 8,000 riyals in an account that grows by 5% each year. Find its value after 2 years.

$8,000 \times 1.05^2 = 8,000 \times 1.1025 = 8,820$ riyals.

- 49** A map has scale 1 : 25,000. Two towns are 8.4 cm apart on the map. Find the actual distance between them in kilometers.

$8.4 \times 25,000 = 210,000$ cm = 2,100 m = 2.1 km.

MCQ — Simplifying & writing ratios

50 Simplify the ratio 42 : 56.

a. 3 : 4 ✓

b. 6 : 8

c. 7 : 8

d. 3 : 8

Divide both terms by their GCD, 14: $42 \div 14 : 56 \div 14 = 3 : 4$.

51 Simplify 18 : 45.

a. 3 : 5

b. 2 : 9

c. 2 : 5 ✓

d. 6 : 15

GCD of 18 and 45 is 9: $18 \div 9 : 45 \div 9 = 2 : 5$.

52 A recipe uses 250 g flour and 2 eggs. Write the ratio of flour (in grams) to eggs in simplest form.

a. 1 : 125

b. 250 : 2

c. 125 : 1 ✓

d. 2 : 250

250 : 2 simplifies by dividing both by 2: 125 : 1.

53 Express $0.75 : 2$ as a ratio of whole numbers in simplest form.

- a. $3 : 4$
- b. $3 : 8$ ✓
- c. $75 : 2$
- d. $75 : 200$

Multiply both terms by 4 to clear the decimal: $3 : 8$.

MCQ — Sharing in a given ratio

54 The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.

- a. 36
- b. 21
- c. 48 ✓
- d. 63

Total parts = $3 + 4 = 7$; each part = $84 \div 7 = 12$. Larger share = $4 \times 12 = 48$.

55 Share 260 SAR in the ratio $2 : 3 : 5$. Find the smallest share.

- a. 78 SAR
- b. 52 SAR ✓
- c. 130 SAR
- d. 65 SAR

Total parts = $2 + 3 + 5 = 10$; each part = $260 \div 10 = 26$. Smallest share = $2 \times 26 = 52$.

56 Two business partners share a profit of 18,000 SAR in the ratio 5 : 7. Find the larger share.

a. 10,500 SAR ✓

b. 9,000 SAR

c. 7,500 SAR

d. 12,857 SAR

Each part = $18,000 \div 12 = 1,500$. Larger share = $7 \times 1,500 = 10,500$ SAR.

57 A line of length 36 cm is divided in the ratio 4 : 5. Find the length of the shorter piece.

a. 16 ✓

b. 4

c. 9

d. 20

Total parts = $4 + 5 = 9$; each part = $36 \div 9 = 4$ cm. Shorter piece = $4 \times 4 = 16$ cm.

58 A class of 40 students has boys and girls in the ratio 3 : 5. How many boys are there?

a. 25

b. 8

c. 24

d. 15 ✓

Total parts = $3 + 5 = 8$; each part = $40 \div 8 = 5$. Boys = $3 \times 5 = 15$.

MCQ — Equivalent ratios & scale

59 If $a : b = 3 : 5$ and $a = 18$, find b .

a. 30 ✓

b. 15

c. 10.8

d. 24

Scale factor $= 18 \div 3 = 6$. $b = 5 \times 6 = 30$.

60 A map has scale $1 : 50,000$. A distance of 3 cm on the map represents how many km in reality?

a. 15 km

b. 1.5 km ✓

c. 150 km

d. 0.15 km

$3 \times 50,000 = 150,000 \text{ cm} = 1,500 \text{ m} = 1.5 \text{ km}$.

61 Which ratio is equivalent to $4 : 9$?

a. $9 : 4$

b. $20 : 45$ ✓

c. $8 : 16$

d. $16 : 37$

$4 : 9$ scaled by 5 gives $20 : 45$.

- 62 In a photo, a model that is 1.8 m tall appears 6 cm high. Using the same scale, how tall (in m) is a building that appears 40 cm high in the photo?
- a. 4.8
 - b. 24
 - c. 7.2
 - d. 12 ✓

Scale: 6 cm represents 1.8 m, so 1 cm represents 0.3 m. $40 \times 0.3 = 12$ m.

MCQ — Percent of a number

- 63 Find 35% of 260.
- a. 86
 - b. 95
 - c. 78
 - d. 91 ✓

$$0.35 \times 260 = 91.$$

- 64 Find 12% of 450.
- a. 62
 - b. 48
 - c. 54 ✓
 - d. 45

$$0.12 \times 450 = 54.$$

65 The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?

- a. 68
- b. 64
- c. 60 ✓
- d. 56

$$75\% \text{ of } 80 = 0.75 \times 80 = 60.$$

66 The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?

- a. Apt. 2
- b. Apt. 1
- c. Apt. 3 ✓
- d. Apt. 4

$$0.08 \times 2,500 = 200 \text{ SAR, matching Apartment 3's rent of 2,500 SAR.}$$

67 150% of a number is 300. Find the number.

- a. 250
- b. 450
- c. 200 ✓
- d. 150

$$1.5 \times x = 300 \Rightarrow x = 300 \div 1.5 = 200.$$

MCQ — Fraction, decimal & percent conversion

68 Write $\frac{7}{20}$ as a percent.

a. 35% ✓

b. 3.5%

c. 7%

d. 70%

$$\frac{7}{20} = \frac{35}{100} = 35\%.$$

69 Write 0.06 as a percent.

a. 60%

b. 6% ✓

c. 0.06%

d. 0.6%

Move the decimal point two places right: $0.06 = 6\%$.

70 Write 85% as a fraction in simplest form.

a. $\frac{17}{20}$ ✓

b. $\frac{8}{5}$

c. $\frac{17}{25}$

d. $\frac{85}{100}$

$$85\% = \frac{85}{100}, \text{ which simplifies (divide by 5) to } \frac{17}{20}.$$

71 Which of these is largest: $\frac{3}{8}$, 0.4, or 37%?

a. 0.4 ✓

b. They are all equal

c. $\frac{3}{8}$

d. 37%

$$\frac{3}{8} = 0.375 = 37.5\%, \text{ and } 37\% = 0.37. \text{ So } 0.4 \text{ is the largest.}$$

MCQ — Percentage increase & decrease

72 Increase 80 by 25%.

- a. 105
- b. 20
- c. 100 ✓
- d. 95

$$80 \times 1.25 = 100.$$

73 Decrease 150 by 12%.

- a. 138
- b. 128
- c. 132 ✓
- d. 18

$$150 \times 0.88 = 132.$$

74 The bar chart shows a salary before and after a raise. What percentage increase does it represent?

- a. 15%
- b. 500%
- c. 20% ✓
- d. 25%

$$\text{Increase} = 6,000 - 5,000 = 1,000. \text{ Percentage increase} = \frac{1,000}{5,000} \times 100\% = 20\%.$$

75 A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 SAR ✓

b. 144 SAR

c. 100 SAR

d. 140 SAR

$$120 = 0.8 \times \text{original} \Rightarrow \text{original} = 120 \div 0.8 = 150 \text{ SAR.}$$

76 Find the percentage change from 40 to 34.

a. 17.6% decrease

b. 15% increase

c. 6% decrease

d. 15% decrease ✓

$$\text{Change} = 34 - 40 = -6. \text{ Percentage change} = \frac{-6}{40} \times 100\% = -15\%, \text{ a 15\% decrease.}$$

MCQ — Finding the original value

77 After a 15% increase, a price is 92 SAR. Find the original price.

a. 80 SAR ✓

b. 78.20 SAR

c. 88 SAR

d. 107 SAR

$$92 = 1.15 \times \text{original} \Rightarrow \text{original} = 92 \div 1.15 = 80 \text{ SAR.}$$

78 After a 30% discount, an item costs 56 SAR. Find the original price.

- a. 72.80 SAR
- b. 80 SAR ✓
- c. 86 SAR
- d. 74.67 SAR

$$56 = 0.7 \times \text{original} \Rightarrow \text{original} = 56 \div 0.7 = 80 \text{ SAR.}$$

79 A population grew by 8% to reach 27,000. Find the population before the growth.

- a. 25,000 ✓
- b. 26,000
- c. 29,160
- d. 24,840

$$27,000 = 1.08 \times \text{original} \Rightarrow \text{original} = 27,000 \div 1.08 = 25,000.$$

80 After including a 15% service charge, a restaurant bill is 138 SAR. Find the bill before the charge.

- a. 158.70 SAR
- b. 120 SAR ✓
- c. 123 SAR
- d. 117.30 SAR

$$138 = 1.15 \times \text{original} \Rightarrow \text{original} = 138 \div 1.15 = 120 \text{ SAR.}$$

MCQ — Successive percentage changes

81 A price of 200 rises 10%, then rises another 10%. Find the final price.

- a. 220
- b. 240
- c. 244
- d. 242 ✓

$$200 \times 1.1 \times 1.1 = 242.$$

82 A price of 500 falls 20%, then rises 20%. Find the final price.

- a. 460
- b. 480 ✓
- c. 500
- d. 520

$$500 \times 0.8 \times 1.2 = 480 \text{ — a rise then fall of the same percentage never returns to the start.}$$

83 The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.

- a. 500
- b. 900
- c. 1,000
- d. 750 ✓

$$1,000 \times 1.5 \times 0.5 = 750.$$

MCQ — Direct proportion

- 84 If 5 workers can build a wall in a task requiring 200 worker-hours total, and pay is directly proportional to hours worked, how much does one worker earn (in SAR) for 8 hours at a rate of 25 SAR/hour?
- a. 40
 - b. 200 ✓
 - c. 225
 - d. 175

$8 \times 25 = 200$ SAR — direct proportion between hours worked and pay.

- 85 12 pens cost 30 SAR. At the same rate, find the cost of 20 pens.
- a. 45 SAR
 - b. 60 SAR
 - c. 40 SAR
 - d. 50 SAR ✓

Cost per pen = $30 \div 12 = 2.5$ SAR. 20 pens cost $20 \times 2.5 = 50$ SAR.

- 86 The graph shows the direct proportion between the number of hours x worked and pay y in SAR, $y = 15x$. How much is earned for 6 hours?
- a. 105
 - b. 90 ✓
 - c. 75
 - d. 21

$y = 15 \times 6 = 90$ SAR.

87 A car uses 8 liters of fuel to travel 96 km. At the same rate, how far can it travel on 20 liters?

- a. 200
- b. 260
- c. 240 ✓
- d. 180

Rate = $96 \div 8 = 12$ km per liter. $20 \times 12 = 240$ km.

MCQ — Inverse proportion

88 6 workers can finish a job in 10 days. How many days would 4 workers take (same rate each)?

- a. 15 ✓
- b. 6.67
- c. 24
- d. 12

Total worker-days = $6 \times 10 = 60$. With 4 workers: $60 \div 4 = 15$ days.

89 If y is inversely proportional to x and $y = 8$ when $x = 3$, find y when $x = 12$.

- a. 0.5
- b. 32
- c. 4
- d. 2 ✓

$xy = 3 \times 8 = 24$ (constant). When $x = 12$: $y = 24 \div 12 = 2$.

90 A tank is filled by 5 identical pumps in 12 hours. How long would 3 of the same pumps take?

- a. 15
- b. 20 ✓
- c. 7.2
- d. 4

Total pump-hours = $5 \times 12 = 60$. With 3 pumps: $60 \div 3 = 20$ hours.

MCQ — Best value & unit price

91 The bar chart shows the price per 100 g for three brands of coffee. Which brand offers the best value?

- a. Brand B ✓
- b. All are equal
- c. Brand C
- d. Brand A

The lowest price per 100 g is the best value — Brand B at 9 SAR/100g beats Brand A (12) and Brand C (11).

92 The bar chart shows the total price of two cereal boxes of different sizes: 400 g and 600 g. Which has the lower price per 100 g?

- a. Cannot be determined
- b. They are equal
- c. The 600 g box ✓
- d. The 400 g box

400 g: $12 \div 4 = 3$ SAR/100g. 600 g: $16.50 \div 6 = 2.75$ SAR/100g. The 600 g box is cheaper per 100 g.

93 Find the unit price (per kg) if 3.5 kg of rice costs 28 SAR.

- a. 7 SAR/kg
- b. 8 SAR/kg ✓**
- c. 9.80 SAR/kg
- d. 3.5 SAR/kg

$$28 \div 3.5 = 8 \text{ SAR per kg.}$$

94 A 2 L bottle of juice costs 18 SAR. Find the price per 100 mL.

- a. 9 SAR
- b. 0.18 SAR
- c. 1.80 SAR
- d. 0.90 SAR ✓**

$$2 \text{ L} = 2,000 \text{ mL. } 18 \div 2,000 \times 100 = 0.90 \text{ SAR per 100 mL.}$$

MCQ — Word problems

95 A car travels 240 km using 20 L of fuel. Find its fuel economy in km per liter.

- a. 14
- b. 10
- c. 20
- d. 12 ✓**

$$240 \div 20 = 12 \text{ km/L.}$$

- 96 The bar chart shows subject preference percentages from a survey of 250 students. How many students prefer math?
- a. 60
 - b. 90
 - c. 110
 - d. 100 ✓

$$0.40 \times 250 = 100 \text{ students.}$$

- 97 A recipe for 4 people needs 2.5 cups of rice. How many cups are needed for 10 people?
- a. 6.25 cups ✓
 - b. 6 cups
 - c. 5 cups
 - d. 25 cups

$$\text{Rice per person} = 2.5 \div 4 = 0.625. \text{ For 10 people: } 10 \times 0.625 = 6.25 \text{ cups.}$$

- 98 A company's revenue rose from 2.4 million to 3.0 million SAR. Find the percentage increase.
- a. 30%
 - b. 25% ✓
 - c. 20%
 - d. 60%

$$\text{Increase} = 3.0 - 2.4 = 0.6 \text{ million. Percentage} = \frac{0.6}{2.4} \times 100\% = 25\%.$$

99 A map scale is 1 : 200,000. Two cities are 8.5 cm apart on the map. Find the real distance in km.

- a. 1.7
- b. 34
- c. 17 ✓
- d. 8.5

$$8.5 \times 200,000 = 1,700,000 \text{ cm} = 17,000 \text{ m} = 17 \text{ km}.$$

Multiple choice — easy

100 Find 15% of 160.

- a. 19
- b. 29
- c. 36
- d. 24 ✓

$$\frac{15}{100} \times 160 = 24.$$

101 Find 80% of 220.

- a. 171
- b. 188
- c. 176 ✓
- d. 181

$$\frac{80}{100} \times 220 = 176.$$

102 Find 150% of 130.

a. 200

b. 207

c. 195 ✓

d. 190

$$\frac{150}{100} \times 130 = 195.$$

103 Find 20% of 370.

a. 69

b. 74 ✓

c. 79

d. 86

$$\frac{20}{100} \times 370 = 74.$$

104 Find 40% of 140.

a. 56 ✓

b. 68

c. 51

d. 61

$$\frac{40}{100} \times 140 = 56.$$

105 Find 60% of 370.

a. 227

b. 222 ✓

c. 217

d. 234

$$\frac{60}{100} \times 370 = 222.$$

106 Find 10% of 400.

a. 52

b. 40 ✓

c. 35

d. 45

$$\frac{10}{100} \times 400 = 40.$$

107 Find 10% of 290.

a. 41

b. 24

c. 29 ✓

d. 34

$$\frac{10}{100} \times 290 = 29.$$

108 Find 20% of 300.

a. 60 ✓

b. 55

c. 72

d. 65

$$\frac{20}{100} \times 300 = 60.$$

109 Find 30% of 400.

a. 115

b. 120 ✓

c. 132

d. 125

$$\frac{30}{100} \times 400 = 120.$$

110 Find 5% of 140.

a. 7 ✓

b. 2

c. 12

d. 19

$$\frac{5}{100} \times 140 = 7.$$

111 Find 5% of 100.

a. 5 ✓

b. 17

c. 10

d. 25

$$\frac{5}{100} \times 100 = 5.$$

112 Find 20% of 340.

a. 68 ✓

b. 80

c. 73

d. 63

$$\frac{20}{100} \times 340 = 68.$$

113 Find 60% of 310.

- a. 198
- b. 191
- c. 181
- d. 186 ✓

$$\frac{60}{100} \times 310 = 186.$$

114 Find 50% of 290.

- a. 145 ✓
- b. 140
- c. 157
- d. 150

$$\frac{50}{100} \times 290 = 145.$$

Multiple choice — medium

115 Increase 340 by 50%.

- a. 519
- b. 506
- c. 510 ✓
- d. 514

$$340 \times 1.5 = 510.$$

116 Decrease 210 by 10%.

- a. 185
- b. 189 ✓
- c. 198
- d. 193

$$210 \times 0.9 = 189.$$

117 Increase 140 by 75%.

- a. 254
- b. 241
- c. 245 ✓
- d. 249

$$140 \times 1.75 = 245.$$

118 Increase 220 by 75%.

- a. 381
- b. 389
- c. 394
- d. 385 ✓

$$220 \times 1.75 = 385.$$

119 Decrease 200 by 75%.

- a. 54
- b. 46
- c. 50 ✓
- d. 59

$$200 \times 0.25 = 50.$$

120 Decrease 160 by 75%.

- a. 44
- b. 49
- c. 36
- d. 40 ✓

$$160 \times 0.25 = 40.$$

121 Increase 310 by 10%.

- a. 350
- b. 345
- c. 341 ✓
- d. 337

$$310 \times 1.1 = 341.$$

122 Increase 380 by 30%.

- a. 498
- b. 503
- c. 490
- d. 494 ✓

$$380 \times 1.3 = 494.$$

123 Decrease 20 by 5%.

- a. 28
- b. 23
- c. 15
- d. 19 ✓

$$20 \times 0.95 = 19.$$

124 Increase 110 by 30%.

- a. 143 ✓
- b. 147
- c. 139
- d. 152

$$110 \times 1.3 = 143.$$

125 Decrease 30 by 30%.

- a. 25
- b. 30
- c. 21 ✓
- d. 17

$$30 \times 0.7 = 21.$$

126 Decrease 340 by 50%.

- a. 166
- b. 179
- c. 174
- d. 170 ✓

$$340 \times 0.5 = 170.$$

127 Decrease 150 by 30%.

- a. 114
- b. 101
- c. 109
- d. 105 ✓

$$150 \times 0.7 = 105.$$

128 Increase 240 by 25%.

- a. 309
- b. 296
- c. 304
- d. 300 ✓

$$240 \times 1.25 = 300.$$

129 Increase 120 by 10%.

- a. 132 ✓
- b. 141
- c. 136
- d. 128

$$120 \times 1.1 = 132.$$

Multiple choice — hard

130 Share 72 in the ratio 5 : 3.

a. 45, 27 ✓

b. 27, 45

c. 54, 27

d. 45, 18

$5 + 3 = 8$ parts, each part $= 72 \div 8 = 9$. Shares: 45, 27.

131 Share 390 among three people in the ratio 7 : 3 : 3.

a. 210, 90, 90 ✓

b. 90, 90, 210

c. 240, 90, 90

d. 210, 90, 60

$7 + 3 + 3 = 13$ parts, each part $= 390 \div 13 = 30$. Shares: 210, 90, 90.

132 Share 300 among three people in the ratio 2 : 4 : 6.

a. 50, 100, 125

b. 50, 100, 150 ✓

c. 150, 100, 50

d. 75, 100, 150

$2 + 4 + 6 = 12$ parts, each part $= 300 \div 12 = 25$. Shares: 50, 100, 150.

133 Share 171 among three people in the ratio 2 : 5 : 2.

a. 38, 95, 19

b. 38, 95, 38 ✓

c. 76, 95, 38

d. 57, 95, 38

$2 + 5 + 2 = 9$ parts, each part $= 171 \div 9 = 19$. Shares: 38, 95, 38.

134 Share 192 among three people in the ratio 3 : 6 : 3.

- a. 64, 96, 48
- b. 48, 96, 48 ✓
- c. 48, 96, 32
- d. 80, 96, 48

$3 + 6 + 3 = 12$ parts, each part $= 192 \div 12 = 16$. Shares: 48, 96, 48.

135 9 kg of apples cost 99 riyals. Find the cost of 7 kg of apples at the same rate.

- a. 88
- b. 66
- c. 77 ✓
- d. 91

Cost per unit $= 99 \div 9 = 11$ riyals. Cost of 7: $11 \times 7 = 77$ riyals.

136 7 liters of paint cost 98 riyals. Find the cost of 5 liters of paint at the same rate.

- a. 80
- b. 56
- c. 70 ✓
- d. 84

Cost per unit $= 98 \div 7 = 14$ riyals. Cost of 5: $14 \times 5 = 70$ riyals.

137 6 meters of fabric cost 84 riyals. Find the cost of 3 meters of fabric at the same rate.

- a. 28
- b. 48
- c. 56
- d. 42 ✓

Cost per unit = $84 \div 6 = 14$ riyals. Cost of 3: $14 \times 3 = 42$ riyals.

138 2 boxes of tiles cost 26 riyals. Find the cost of 5 boxes of tiles at the same rate.

- a. 65 ✓
- b. 78
- c. 75
- d. 52

Cost per unit = $26 \div 2 = 13$ riyals. Cost of 5: $13 \times 5 = 65$ riyals.

139 6 kg of rice cost 30 riyals. Find the cost of 2 kg of rice at the same rate.

- a. 10 ✓
- b. 5
- c. 14
- d. 15

Cost per unit = $30 \div 6 = 5$ riyals. Cost of 2: $5 \times 2 = 10$ riyals.

140 After a 50% decrease, a quantity is now 145. Find the original value.

- a. 300
- b. 280
- c. 290 ✓
- d. 145

Original $\times 0.5 = 145$, so original = $145 \div 0.5 = 290$.

141 After a 50% increase, a quantity is now 90. Find the original value.

a. 60 ✓

b. 70

c. 50

d. 90

Original $\times 1.5 = 90$, so original $= 90 \div 1.5 = 60$.

142 After a 30% decrease, a quantity is now 63. Find the original value.

a. 80

b. 100

c. 63

d. 90 ✓

Original $\times 0.7 = 63$, so original $= 63 \div 0.7 = 90$.

143 After a 40% increase, a quantity is now 532. Find the original value.

a. 390

b. 380 ✓

c. 370

d. 532

Original $\times 1.4 = 532$, so original $= 532 \div 1.4 = 380$.

144 After a 10% decrease, a quantity is now 360. Find the original value.

a. 400 ✓

b. 390

c. 360

d. 410

Original $\times 0.9 = 360$, so original $= 360 \div 0.9 = 400$.